Project Title

“Revenue Growth and Sales Performance Intelligence Dashboard”

Turning Sales & Operational Data into Strategic Revenue Decisions

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Report Prepared By

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Tools & Technologies Used

Power BI
Power Query Editor (PQE)
DAX
Star Schema Data Modeling

ABSTRACT

The project titled **“Revenue Growth and Sales Performance Intelligence Dashboard”** focuses on transforming raw transactional and operational sales data into a structured, decision-support analytics system using Power BI.

In today’s competitive business environment, organizations require more than descriptive reporting — they need analytical insights that identify revenue drivers, operational inefficiencies, and growth opportunities. This project integrates financial metrics, product-level intelligence, seller performance data, and customer satisfaction indicators to provide a comprehensive view of business performance.

Using Power Query Editor for data cleaning and transformation, DAX for KPI and measure creation, and Star Schema data modeling for optimized relationships, a multi-page interactive dashboard was developed. The dashboard covers:

- Executive Revenue Overview
- Product & Category Intelligence
- Seller & Operational Performance
- Root Cause & Revenue Driver Analysis
- Insights, Recommendations & Limitations

The analysis reveals key findings such as revenue concentration across specific states and sellers, freight cost impact on margin efficiency, and strong correlation between delivery delays and negative customer reviews.

The final output is an executive-ready, analytics-driven business intelligence solution that supports data-driven revenue growth strategies and operational improvements.

DECLARATION

I hereby declare that the project report titled **“Revenue Growth and Sales Performance Intelligence Dashboard”** is a bonafide record of work carried out by me as part of my academic/portfolio project work.

This project has been developed using Power BI, Power Query Editor (PQE), DAX, and Star Schema Data Modeling techniques. The analysis, dashboard design, business insights, and recommendations presented in this report are the result of my independent effort and understanding of business intelligence concepts.

All data transformations, modeling, and analytical measures were performed by me, and the report has been prepared solely for academic and portfolio demonstration purposes.

I further declare that this project has not been submitted to any other institution or organization for the award of any degree or certification.

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Date: 3/3/2026
Place: Bangalore

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1. Project Overview

This project analyzes sales, product, seller, and operational data to uncover actionable insights that support sustainable revenue growth and operational efficiency.

The dashboard is designed as a multi-layer decision-support system that answers critical business questions such as:

- Which states and sellers drive the highest revenue?
- How concentrated is revenue across categories and sellers?
- How does delivery speed impact customer satisfaction?
- What operational factors drive negative reviews?
- Where are the biggest revenue growth opportunities?

The primary objective of this project is:

To transform raw transactional and operational data into a structured, executive-level intelligence dashboard that enables data-driven revenue growth strategies.

Unlike simple descriptive dashboards, this solution integrates financial, operational, and customer experience metrics to identify root causes of performance gaps.

2. Dataset Summary

The dataset represents a multi-dimensional e-commerce sales environment containing:

- ~100,000 orders
- Revenue data across multiple states
- Product categories and SKUs
- Seller-level performance data
- Delivery timelines
- Customer review ratings

Key Data Domains:

- Financial Metrics: Revenue, Order Value, Freight Cost
- Geographic Metrics: Customer State, Seller State
- Product Data: Category, Product ID
- Seller Data: Seller ID
- Operational Metrics: Delivery Days
- Customer Experience: Review Score, Review Category

Data Constraints:

- No direct profit column (profit approximated using revenue vs freight analysis)
- No marketing spend data
- No customer lifetime tracking
- No detailed cancellation or return reason codes

3. Data Preparation & Modeling

Before dashboard development, the data underwent structured transformation and modeling.

Data Cleaning (Power Query Editor)

- Removed duplicate records
- Standardized date formats
- Validated state and category naming consistency
- Created delivery duration metric (Order Delivered Date – Order Purchase Date)
- Categorized review scores into Positive / Neutral / Negative

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Feature Engineering (DAX)

Created calculated measures including:

- Total Revenue
- Total Orders
- Average Order Value (AOV)
- YoY Revenue Growth %
- Positive Review %
- Average Delivery Days
- Freight-to-Revenue Ratio
- Cumulative Seller Revenue % (Pareto Analysis)

These metrics enabled executive-level KPI tracking.

Data Modeling (Star Schema)

A Star Schema model was implemented for optimal performance and scalability.

Fact Table:

- Orders / Sales Fact (Revenue, Freight, Delivery Days, Review Score)

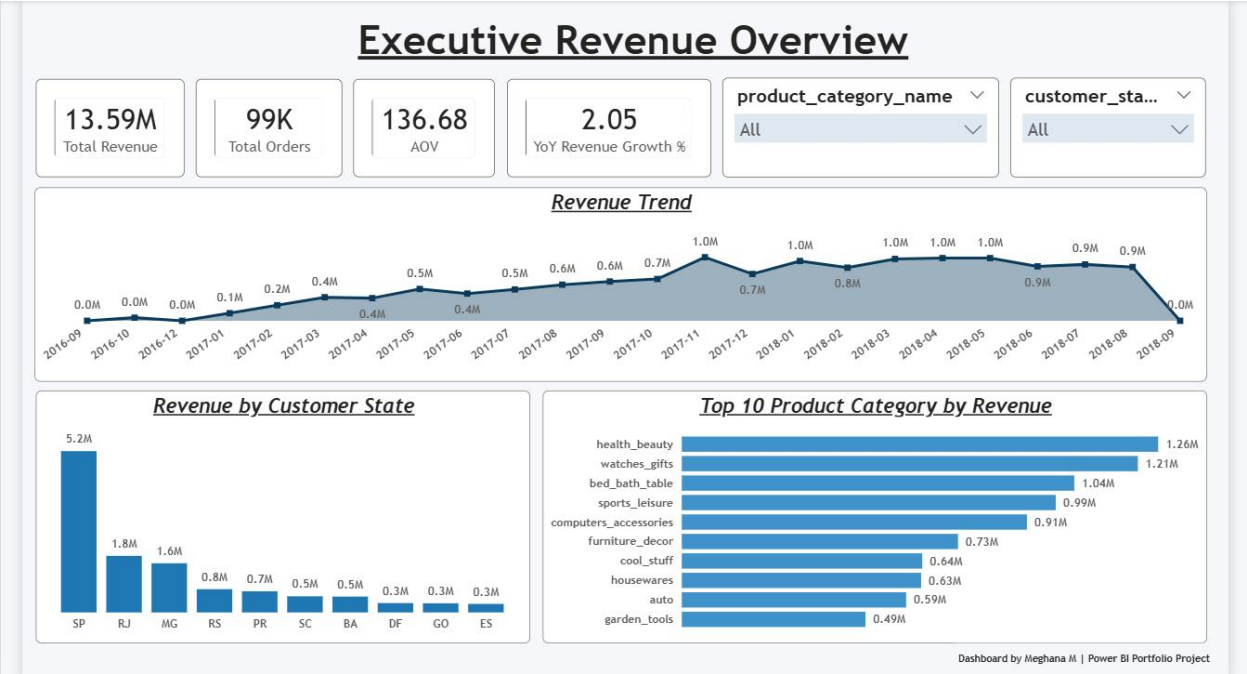
Dimension Tables:

- Date Dimension
- Customer State
- Seller
- Product
- Category

This structure enabled efficient slicing by state, category, seller, and time without performance degradation.

4. Executive & Sales Performance Analysis

4.1 Executive Revenue Overview



This page provides a macro-level view of business performance.

KPI Summary:

- Total Revenue: 13.59M
- Total Orders: 99K
- AOV: 136.68
- YoY Growth: 2.05%

Key Analysis Areas:

1. Revenue Trend Over Time
   Revenue peaked in late 2017 followed by gradual stabilization and minor decline.

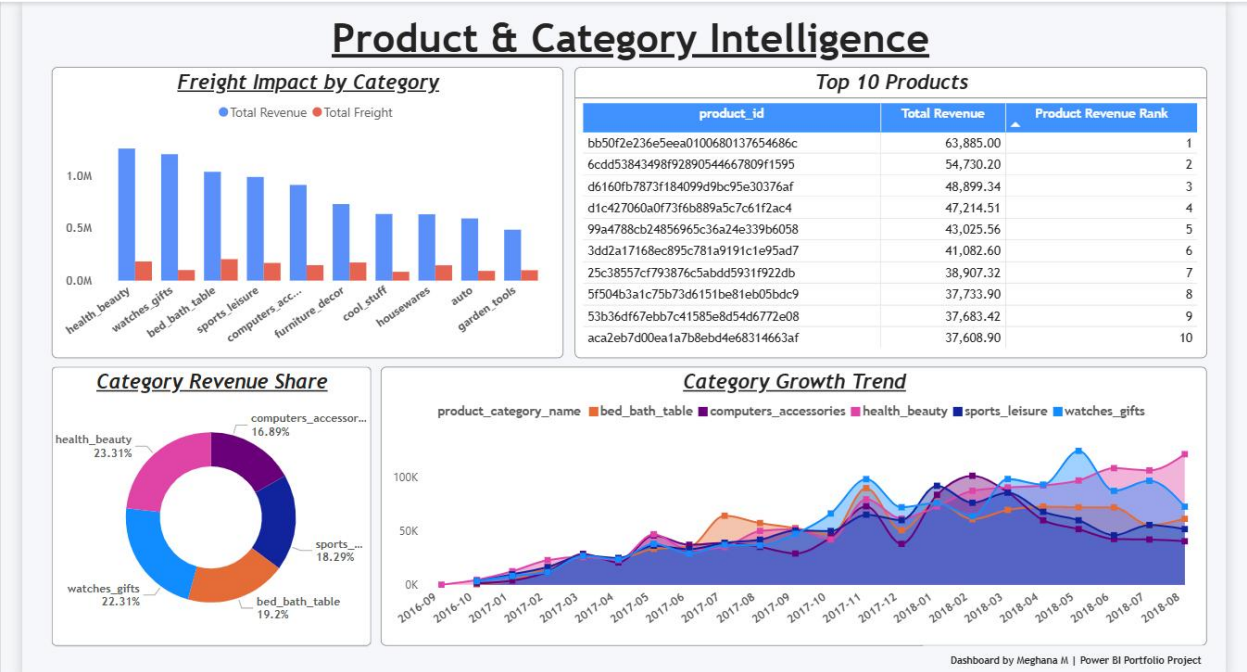
This signals potential market saturation or operational bottlenecks.

2. Revenue by State
   SP, RJ, and MG contribute the largest share of revenue.
   Revenue is geographically concentrated.

3. Top 10 Product Categories
   Categories such as health_beauty, watches_gifts, and bed_bath_table dominate revenue contribution.

Business Insight:
Growth is stable but not accelerating — strategic expansion and operational improvements are required.

4.2 Product & Category Intelligence



This page analyzes revenue drivers at product and category levels.

Key Findings:

1. Category Revenue Distribution
   Top 3 categories contribute ~60% of total revenue → high dependency risk.

2. Freight Impact Analysis
   Categories like bed_bath_table and sports_lesiure show high freight cost ratios, reducing margin efficiency.

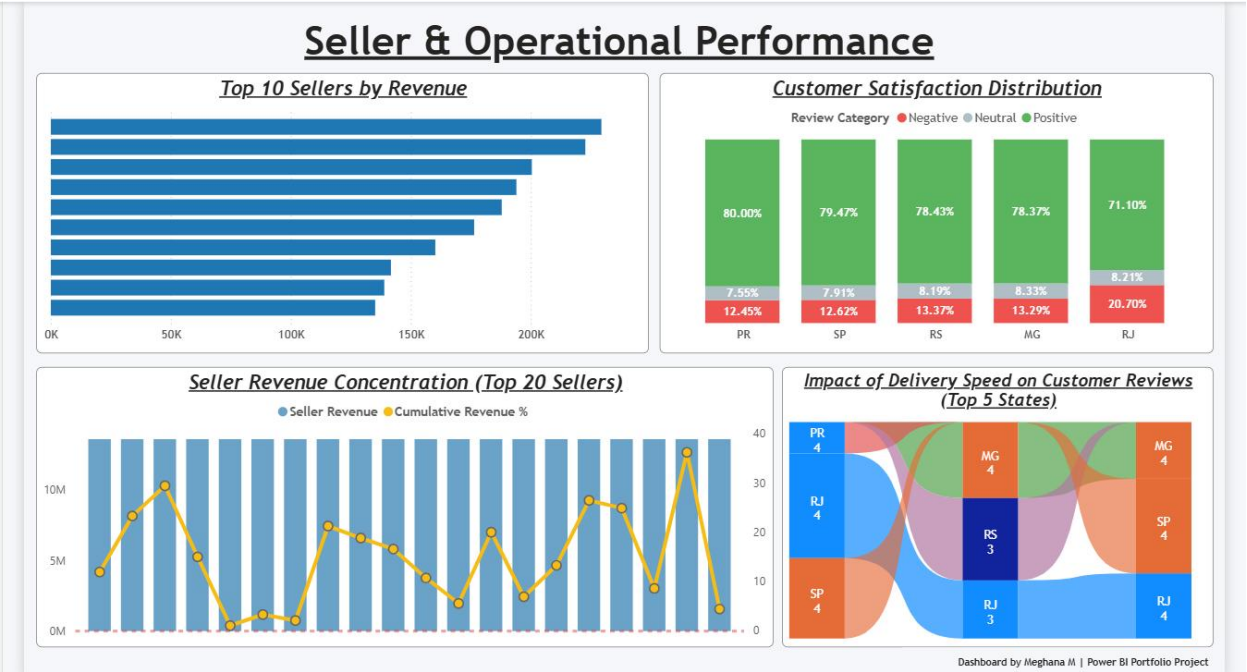
3. Top Products by Revenue
   Revenue is further concentrated at SKU level within specific categories.

4. Category Growth Trends
   Significant spikes observed mid-2017; later growth stabilizes.

Business Insight:

Revenue concentration + high freight categories may impact long-term profitability.

4.3 Seller & Operational Performance



This section evaluates seller performance and operational efficiency.

Key Findings:

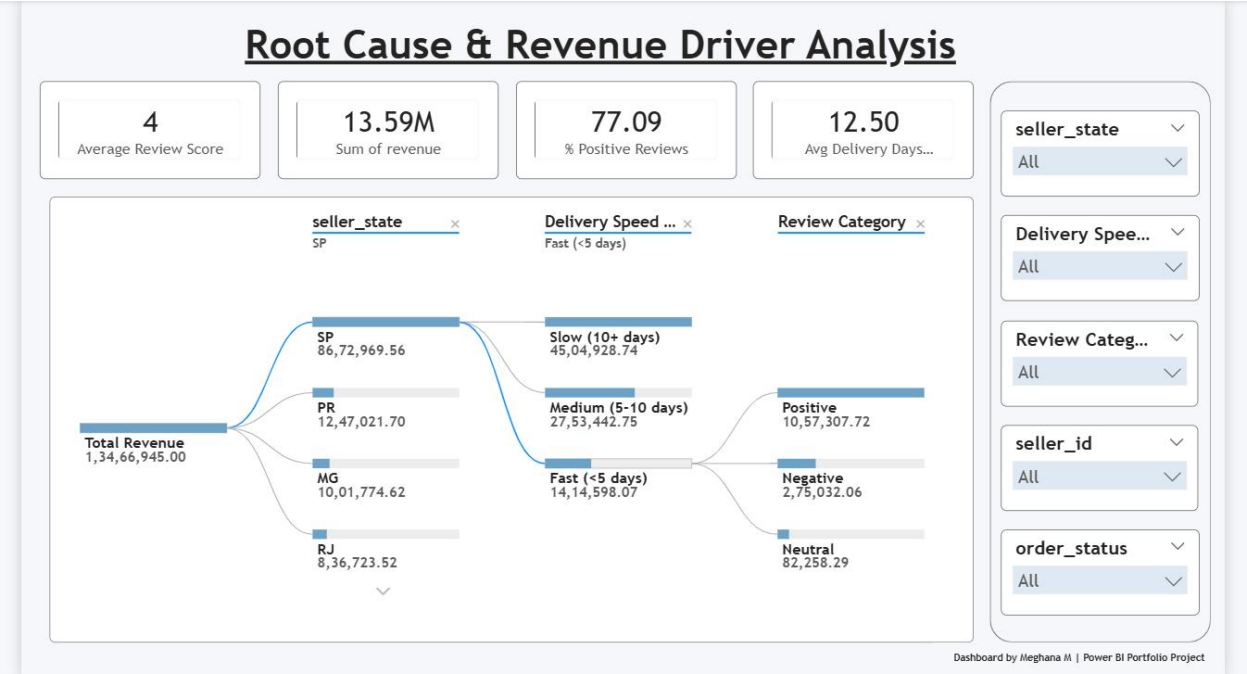
1. Revenue Concentration (Pareto Pattern)
   Top 5–10 sellers generate majority of revenue.

2. Customer Satisfaction by State
   Some states (e.g., RJ) show higher negative review ratios.

3. Delivery Speed vs Review Sentiment
   Fast deliveries → Higher positive reviews
   Slow deliveries (10+ days) → Surge in negative reviews

Business Insight:
Delivery performance directly impacts customer satisfaction and potentially revenue sustainability.

4.4 Root Cause & Revenue Driver Analysis



This page provides deeper diagnostic analysis.

Decomposition Analysis:

Revenue and review performance broken down by:

- Seller State
- Delivery Speed
- Review Category

Key Finding:

Slow deliveries (10+ days) are strongly linked to:

- Higher negative reviews
- Revenue leakage risk
- Concentration in specific states

Business Insight:
Operational delays are the primary root cause of satisfaction decline.

5. Power BI Dashboard Design

The dashboard was structured as a 5-page decision-support system:

- Page 1: Executive Overview
- Page 2: Product & Category Intelligence
- Page 3: Seller & Operational Performance
- Page 4: Root Cause Analysis
- Page 5: Insights, Recommendations & Limitations

Design Features:

- KPI cards for executive clarity
- Interactive slicers (state, category, seller)
- Decomposition tree for drill-down diagnostics
- Sankey chart for delivery vs review analysis
- Tooltip page for micro-level state review metrics
- Consistent color hierarchy and business-friendly layout

The design prioritizes decision-making over visual complexity.

6. Key Business Insights

1. Revenue is geographically concentrated (SP, RJ, MG).
2. Top 3 categories contribute ~60% of total revenue.
3. Revenue heavily depends on top sellers (Pareto distribution).
4. Slow delivery (10+ days) significantly increases negative reviews.
5. High freight categories reduce margin efficiency.
6. Growth after 2017 shows signs of stabilization, not acceleration.

7. Actionable Recommendations

1. Reduce Slow Deliveries
   Prioritize logistics optimization in states with high delay rates.

2. Optimize Freight for Heavy Categories
   Negotiate logistics rates or adjust pricing strategy.

3. Diversify Revenue Base
   Promote mid-tier sellers to reduce dependency on top 5 sellers.

4. Target High-Performing States
   Increase marketing in SP and MG to maximize ROI.

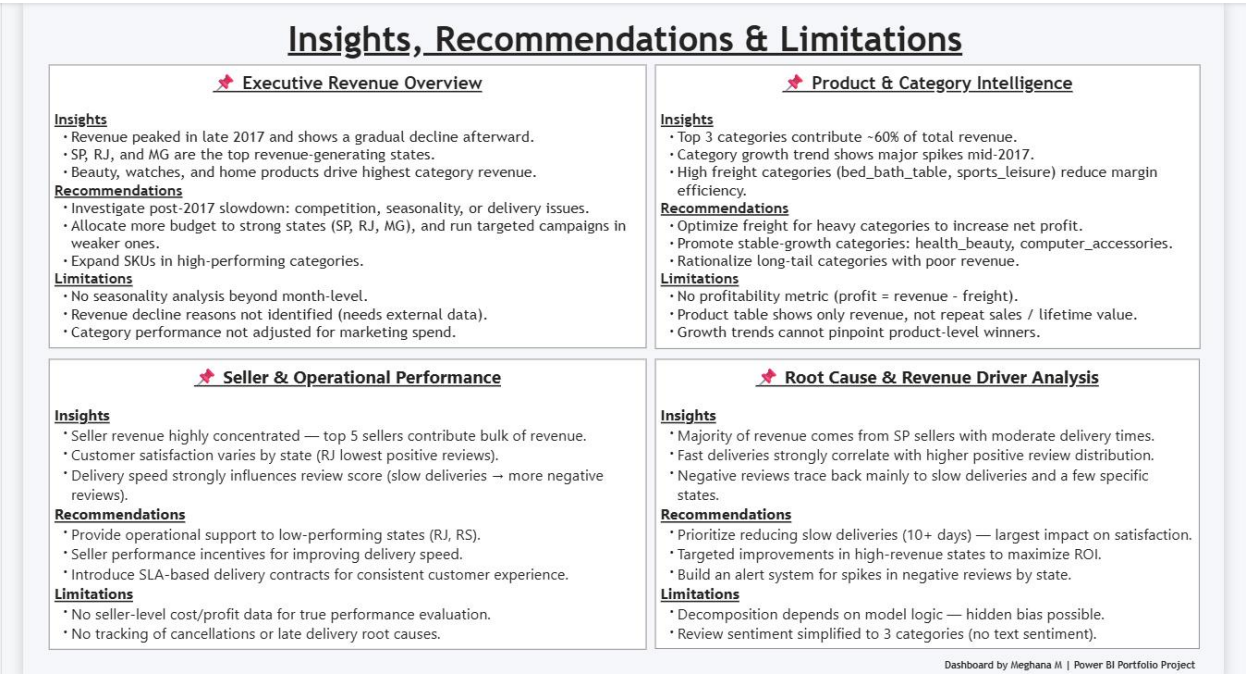
5. Expand High-Growth Categories
   Scale SKUs in stable-growth categories like beleza_saude.

6. Implement Operational Alert System
   Monitor spikes in negative reviews by state in real-time.

8. Limitations & Assumptions

- No direct profitability metric (profit not available).
- No marketing spend data to analyze ROI.
- No customer lifetime value tracking.
- Review sentiment categorized into 3 buckets (no text NLP).
- Root cause insights based on delivery duration proxy.

All limitations are transparently acknowledged to maintain analytical credibility.



Insights, Recommendations & Limitations

📌 Executive Revenue Overview

Insights
- Revenue peaked in late 2017 and shows a gradual decline afterward.
- SP, RJ, and MG are the top revenue-generating states.
- Beauty, watches, and home products drive highest category revenue.

Recommendations
- Investigate post-2017 slowdown: competition, seasonality, or delivery issues.
- Allocate more budget to strong states (SP, RJ, MG), and run targeted campaigns in weaker ones.
- Expand SKUs in high-performing categories.

Limitations
- No seasonality analysis beyond month-level.
- Revenue decline reasons not identified (needs external data).
- Category performance not adjusted for marketing spend.

📌 Product & Category Intelligence

Insights
- Top 3 categories contribute ~60% of total revenue.
- Category growth trend shows major spikes mid-2017.
- High freight categories (bed_bath_table, sports_leisure) reduce margin efficiency.

Recommendations
- Optimize freight for heavy categories to increase net profit.
- Promote stable-growth categories: health_beauty, computer_accessories.
- Rationalize long-tail categories with poor revenue.

Limitations
- No profitability metric (profit = revenue - freight).
- Product table shows only revenue, not repeat sales / lifetime value.
- Growth trends cannot pinpoint product-level winners.

📌 Seller & Operational Performance

Insights
- Seller revenue highly concentrated — top 5 sellers contribute bulk of revenue.
- Customer satisfaction varies by state (RJ lowest positive reviews).
- Delivery speed strongly influences review score (slow deliveries → more negative reviews).

Recommendations
- Provide operational support to low-performing states (RJ, RS).
- Seller performance incentives for improving delivery speed.
- Introduce SLA-based delivery contracts for consistent customer experience.

Limitations
- No seller-level cost/profit data for true performance evaluation.
- No tracking of cancellations or late delivery root causes.

📌 Root Cause & Revenue Driver Analysis

Insights
- Majority of revenue comes from SP sellers with moderate delivery times.
- Fast deliveries strongly correlate with higher positive review distribution.
- Negative reviews trace back mainly to slow deliveries and a few specific states.

Recommendations
- Prioritize reducing slow deliveries (10+ days) — largest impact on satisfaction.
- Targeted improvements in high-revenue states to maximize ROI.
- Build an alert system for spikes in negative reviews by state.

Limitations
- Decomposition depends on model logic — hidden bias possible.
- Review sentiment simplified to 3 categories (no text sentiment).

Dashboard by Meghana M | Power BI Portfolio Project

9. Conclusion

The Revenue Growth and Sales Performance Intelligence Dashboard transforms multi-dimensional sales and operational data into a structured executive decision-support system.

This project demonstrates:

- Advanced data modeling using Star Schema
- Strategic KPI development using DAX
- Operational root cause diagnostics
- Revenue concentration risk identification
- Business-aligned storytelling through dashboards

Rather than simply reporting sales numbers, this dashboard identifies:

- Revenue vulnerabilities
- Operational inefficiencies
- Margin pressure areas
- Growth expansion opportunities

The final output is a business-ready analytics solution designed for executive, operational, and strategic stakeholders.